Appendix 1 *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

Urtzi Enriquez-Urzelai[1*], Gioele Pascucci[1,2], and Lumír Gvoždík[1]

[1]Czech Academy of Sciences, Institute of Vertebrate Biology, Brno, Czech Republic
[2]Department of Botany and Zoology, Faculty of Science, Masaryk University, Brno, Czech Republic
[*]Corresponding author: uenriquez@proton.me

Supplementary Materials and Methods

1 The standard DEB model

Dynamic Energy Budget (DEB) theory (Kooijman, 2010; Sousa et al., 2008) describes the acquisition, storage, and allocation of energy and mass throughout the life cycle. The standard DEB model (Marques et al., 2018) represents an organism by three state variables: structural volume V, reserve energy E, and cumulative energy invested in maturation E_H (plus a reproduction buffer E_R in adults). Energy enters the organism as food, is assimilated into reserve at a rate proportional to surface area, and is subsequently mobilised (catabolised) from the reserve to fuel five downstream processes: somatic maintenance, growth in structure, maturity maintenance, maturation (in juveniles), and reproduction (in adults). Reproduction energy accumulates in a buffer and is periodically converted into eggs. Life-history transitions occur at species-specific maturity thresholds: e.g., birth at E_H^b and sexual maturity at E_H^p (Kooijman, 2010; Marques et al., 2018; Oliveira et al., 2026).

The behaviour of the standard model is governed by 12 primary parameters (Kooijman, 2010). The maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ controls the rate of energy acquisition and, through covariation rules, scales with maximum body size via the dimensionless zoom factor z. A larger z implies a larger body, a proportionally higher assimilation capacity, and a greater energetic investment at life-history transitions (Sousa et al., 2008). Once assimilated, reserves are mobilised at a rate set by the energy conductance $\dot{v}$ and following the κ-rule: a fixed fraction κ of mobilised reserve is allocated to somatic maintenance, whose cost per unit structural volume is $[\dot{p}_M]$, and to growth, at a specific cost of $[E_G]$ per unit of new structure; the remainder $1-\kappa$ supports maturity maintenance (at a rate set by $\dot{k}_J$) and maturation in juveniles or reproduction in adults, with eggs produced at an efficiency κ_R. The rates of feeding and assimilation are further determined by the maximum specific searching rate $\{\dot{F}_m\}$, the assimilation efficiency κ_X, and the surface-specific maintenance cost $\{\dot{p}_T\}$, which is assumed to be zero for ectotherms. The cumulative energetic investments at which organisms reach birth and sexual maturity are defined by the thresholds E_H^b and E_H^p. Auxiliary parameters include the shape coefficient δ_M, which allows to convert shape-specific length measures to lengths relevant in the DEB context, and the acceleration of ageing $\ddot{h}_a$. Sex-specific parameters can also be used to account for between-sex differences, for instance in male puberty ($E_{H,m}^p$) and body-size dimorphism (z_m), and, in species with a larval stage, the standard model can be extended with an additional maturity threshold at metamorphosis, E_H^j. Because each parameter quantifies the rate or efficiency of a specific physiological process, the standard DEB model

provides a coherent framework for interpreting individual variation in terms of energetic traits with distinct ecological meaning (Kooijman et al., 2021).

2 DEB parameter estimation

We fitted the standard DEB model for *I. alpestris* by inverse modelling, using the covariation method (Lika et al., 2011; Marques et al., 2018) as implemented in the freely available AmPtool and DEBtool_M packages for MATLAB (The MathWorks Inc.). Following standard practice in this method, each data set was assigned the temperature at which it was obtained, and rates were corrected for temperature with the Arrhenius relationship. Model goodness-of-fit was assessed with the mean relative error (MRE) and the symmetric mean squared error (SMSE) (Marques et al., 2018).

For the population-average parameterisation, the model was fitted simultaneously to the species-level life-history information and the complete set of longitudinal growth and metabolic observations from all 37 individuals. Individual observations were retained as separate data points rather than averaged across individuals or ages, and male and female observations were fitted jointly within the same parameter-estimation procedure, while allowing sex-specific parameters. Of the 12 primary parameters, we estimated seven: the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$, via the zoom factor z; the energy conductance $\dot{v}$; the allocation fraction κ; the volume-specific somatic maintenance $[\dot{p}_M]$; the specific cost of structure $[E_G]$; and the maturity thresholds at birth E_H^b and puberty E_H^p. The remaining five ($\{\dot{F}_m\}$, κ_X, κ_R, $\{\dot{p}_T\}$ and $\dot{k}_J$) were fixed to standard values from the DEB literature (Kearney et al., 2013; Kooijman, 2010). We also estimated the maturity threshold at metamorphosis E_H^j, which marks the timing and size of metamorphosis without changing the physiology of the model (i.e., no metabolic acceleration was assumed); the male puberty threshold $E_{H,m}^p$ and the male zoom factor z_m, which account for sexual size dimorphism; the shape coefficient δ_M; and the Weibull ageing acceleration $\ddot{h}_a$. The Arrhenius temperature T_A and all chemical parameters were fixed at standard values. Together, these estimated parameters define a model representation of the species' energetic phenotype, with each parameter controlling a distinct aspect of energy acquisition, allocation, or expenditure. Model parameters are given in **Table S2.1**, and the full parameter set and goodness of fit can be found at https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/entries_web/Ichthyosaura_alpestris/Ichthyosaura_alpestris_res.html.

Individual-level parameters were obtained by re-estimating three parameters independently for each individual (the sex-specific zoom factor, z for females and z_m for males; the energy conductance $\dot{v}$; and the volume-specific somatic maintenance $[\dot{p}_M]$), while all remaining parameters, including κ, $[E_G]$ and the maturity thresholds, were held at their population-average values. Each estimation used the individual's own longitudinal growth and metabolic trajectories as univariate data, with population-average estimates serving as starting values, and each individual was assigned the sex-appropriate puberty threshold (E_H^p for females, $E_{H,m}^p$ for males).

3 Microclimate and ectotherm simulations

We reconstructed microclimatic conditions using NicheMapR's microclimate model, as implemented in the `micro_global()` function (Kearney & Porter, 2017; Kearney et al., 2014), which uses monthly climatological means as forcing environmental variables.

The model uses monthly climatological data (maximum and minimum air temperature, precipitation, cloud cover, relative humidity, wind speed and number of wet days) from the global 10 arc-minute climatology of New et al. (2002), together with elevation, slope, aspect and soil properties, to estimate hourly aboveground air temperature, solar and infrared radiation, wind speed and relative humidity at the animal's height, as well as soil temperature and moisture profiles at ten depths (Kearney & Porter, 2017; Kearney et al., 2014). Microclimate simulations were run for 22 years with the moisture and snow routines enabled (Kearney, 2020; Kearney & Maino, 2018). The first two years were excluded as spin-up to ensure stabilised snow and soil moisture dynamics before organismal simulations, resulting in simulations of up to 20 years (i.e., the maximum known lifespan of the species). We ran the microclimate simulations in two shading conditions: fully unshaded (0% shade) and deep shade (90% shade).

To estimate *I. alpestris* performance and simulate its life cycle with coupled biophysical and DEB models, we used the `ectotherm()` function of NicheMapR (Kearney & Porter, 2020). We parameterised the ectotherm model for a terrestrial alpine newt, with 90% of the skin surface acting as a free water exchanger to account for evaporative water loss (Enriquez-Urzelai & Gvoždík, 2024). Food availability was held constant throughout the simulations, so that the amount of energy acquired depended on the time available for foraging. Individuals were assumed to be nocturnal and crepuscular, but juveniles were also allowed diurnal activity, reflecting the need to acquire energy reserves before their first winter (Enriquez-Urzelai et al., 2020). Animals retreated underground during daylight hours (in the case of adults), and whenever body temperature fell outside the voluntary thermal limits (i.e., the lower and upper body temperatures within which an individual is assumed to remain behaviourally active; $VT_{min} = 4\,°C$, $VT_{max} = 20\,°C$), or relative humidity aboveground was lower than 65%. This relative humidity value corresponds well with the lowest relative humidity selected by amphibians at low temperatures (Bonnefond et al., 2020; Giacometti & Tattersall, 2025). When burrowed, animals selected a depth (from 2.5 cm up to 2 m) that prevented them from reaching critical thermal limits (i.e., temperatures at which physiological performance fails; $CT_{min} = -2\,°C$ and $CT_{max} = 36\,°C$) (Enriquez-Urzelai & Gvoždík, 2024). The preferred body temperature was set to 17.3 °C (Gvoždík & Kristín, 2017). Although amphibians are not as accurate thermoregulators as non-avian reptiles, we allowed individuals to behaviourally select shaded microhabitats by adjusting the amount of shade needed (from 0% to 90% of shade) to stay within their preferred temperature range (Enriquez-Urzelai et al., 2019).

In the case of females (both the population-average and individual-level parameterisations), we simulated reproduction using the batch reproductive model (Pecquerie et al., 2009): females stored energy in the reproductive buffer until it was enough to produce a clutch (200–600 eggs) during the breeding season. In males, we simulated continuous depletion of the reproductive buffer during the breeding season (representing energy invested in sperm, courtship, etc.), so no energy allocated to reproduction was stored. The breeding season occurred between the spring equinox and the summer solstice. We ran the simulations from the juvenile stage (starting on 15 August, when juveniles are present in the field), for the total length of the microclimate simulation, until the individual died.

4 Jensen bias and variability metrics

Positive values of Jensen bias ($J = P_{\text{pop}} - \overline{P}_{\text{ind}}$) indicate that the population-average parameterisation overpredicts performance relative to the mean of individuals, whereas negative values indicate underprediction. To avoid interpreting numerical noise as bias, we considered grid cells with $|J| \leq 10^{-10}$ as showing no bias, and, when quantifying the proportion of grid cells in which absolute Jensen bias increased, decreased or remained unchanged under warming, we considered changes with $|\Delta|J|| \leq 10^{-10}$ as unchanged. To compare the magnitude of bias among metrics measured in different units, we computed the normalised absolute Jensen bias for each metric and sex under the current climate, as the mean $|J|$ across grid cells divided by the mean $|\overline{P}_{\text{ind}}|$ across grid cells ($\times 100$).

The coefficient of variation among individual-level predictions was computed for each sex (18 females and 19 males), metric and grid cell as $\text{CV} = SD/(|\overline{P}_{\text{ind}}| + \varepsilon)$, where the small constant $\varepsilon = 10^{-10}$ avoids division by zero. To summarise its geographic pattern, we grouped grid cells into southern (<40° N), central (40°–60° N) and northern (>60° N) latitudinal bands and compared the median CV of each marginal band with that of the central band.

5 Jensen bias along the temperature gradient

Mean annual air temperature ($\overline{T}_a$) was calculated for each grid cell from the global 10 arc-minute climatology of New et al. (2002) by averaging the monthly means of the daily maximum and minimum air temperatures. The change in absolute Jensen bias under +4 °C warming ($\Delta|J|$) was rasterised at approximately 0.16° resolution for each of the 12 performance metrics and summarised as binned means in 1 °C $\overline{T}_a$ intervals, retaining only bins containing at least 10 grid cells. To describe the overall trend along the $\overline{T}_a$ gradient, we fitted generalised additive models (GAMs) with the `mgcv` R package (Wood, 2025) to the cell-level $\Delta|J|$ values, separately for each metric and sex. Models assumed a Gaussian error distribution with identity link and included a thin-plate regression spline smooth of $\overline{T}_a$ (basis dimension $k = 12$), with smoothing parameters estimated by restricted maximum likelihood (REML) and an additional shrinkage penalty (`select = TRUE`). Predictions were evaluated at 0.1 °C intervals along the $\overline{T}_a$ gradient. Because model outputs are deterministic and neighbouring grid cells are spatially autocorrelated, we used GAMs as descriptive smoothers rather than for statistical inference, and we did not report confidence intervals.

6 Comparison of population-average and individual-level forecasts

For each of the three life-history traits (lifespan, maximum body mass and lifetime fecundity, or reproductive energy for males), grid cell and climate scenario, we extracted the population-average prediction (P_{pop}) and the corresponding mean individual-level prediction ($\overline{P}_{\text{ind}}$). Comparisons included both the absolute predictions under the current climate and the projected changes under warming, expressed relative to the current climate ($P_{\text{future}} - P_{\text{current}}$) for each physiological parameterisation. For visualisation, we normalised the projected changes to range between -1 and 1 by dividing each grid cell's value by the maximum absolute change within each trait. We then calculated, for each trait, sex and

scenario, the percentage of grid cells in which the population-average model under- or overpredicted the individual-level mean, and the percentage of grid cells in which both parameterisations projected the same direction of change under warming.

We also computed the relative bias for each trait, grid cell and scenario as the percentage deviation of the population-average prediction from the individual-level mean, $(P_{\text{pop}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$, so that positive values indicate overestimation by the population-average model, negative values underestimation, and values near zero close agreement. Relative bias is undefined in grid cells where the individual-level mean is zero, and these cells were excluded, and it can become very large where the individual-level mean is close to zero. To limit the influence of such near-zero denominators, we summarised the distributions of relative bias across the species' range using the median relative bias and the 95th percentile of the absolute relative bias (i.e., the magnitude of relative bias below which 95% of grid-cell estimates fall).

7 References

Bonnefond, A., Courtois, E. A., Sueur, J., Sugai, L. S. M., & Llusia, D. (2020). Climatic breadth of calling behaviour in two widespread Neotropical frogs: Insights from humidity extremes. *Global Change Biology*, *26*(10), 5431–5446. https://doi.org/10.1111/gcb.15266

Enriquez-Urzelai, U., & Gvoždík, L. (2024). Impacts of behaviour and acclimation of metabolic rate on energetics in sheltered ectotherms: A climate change perspective. *Proceedings of the Royal Society B: Biological Sciences*, *291*(2017), 20232152. https://doi.org/10.1098/rspb.2023.2152

Enriquez-Urzelai, U., Kearney, M. R., Nicieza, A. G., & Tingley, R. (2019). Integrating mechanistic and correlative niche models to unravel range-limiting processes in a temperate amphibian. *Global Change Biology*, *25*(8), 2633–2647. https://doi.org/10.1111/gcb.14673

Enriquez-Urzelai, U., Tingley, R., Kearney, M. R., Sacco, M., Palacio, A. S., Tejedo, M., & Nicieza, A. G. (2020). The roles of acclimation and behaviour in buffering climate change impacts along elevational gradients (K. Marske, Ed.). *Journal of Animal Ecology*, *89*(7), 1722–1734. https://doi.org/10.1111/1365-2656.13222

Giacometti, D., & Tattersall, G. J. (2025). Behavioural evidence of a humidistat: A temperature-compensating mechanism of hydroregulation in spotted salamanders. *Journal of Experimental Biology*, *228*(10), jeb250297. https://doi.org/10.1242/jeb.250297

Gvoždík, L., & Kristín, P. (2017). Economic thermoregulatory response explains mismatch between thermal physiology and behavior in newts. *Journal of Experimental Biology*, *220*(6), jeb.145573. https://doi.org/10.1242/jeb.145573

Kearney, M. R. (2020). How will snow alter exposure of organisms to cold stress under climate warming? (A. Guisan, Ed.). *Global Ecology and Biogeography*, *29*(7), 1246–1256. https://doi.org/10.1111/geb.13100

Kearney, M. R., & Maino, J. L. (2018). Can next-generation soil data products improve soil moisture modelling at the continental scale? An assessment using a new microclimate package for the R programming environment. *Journal of Hydrology*, *561*, 662–673. https://doi.org/10.1016/j.jhydrol.2018.04.040

Kearney, M. R., & Porter, W. P. (2017). NicheMapR – an R package for biophysical modelling: The microclimate model. *Ecography*, *40*(5), 664–674. https://doi.org/10.1111/ecog.02360

Kearney, M. R., & Porter, W. P. (2020). NicheMapR – an R package for biophysical modelling: The ectotherm and dynamic energy budget models. *Ecography*, *43*(1), 85–96. https://doi.org/10.1111/ecog.04680

Kearney, M. R., Shamakhy, A., Tingley, R., Karoly, D. J., Hoffmann, A. A., Briggs, P. R., & Porter, W. P. (2014). Microclimate modelling at macro scales: A test of a general microclimate model integrated with gridded continental-scale soil and weather data (J. Travis, Ed.). *Methods in Ecology and Evolution*, *5*(3), 273–286. https://doi.org/10.1111/2041-210X.12148

Kearney, M. R., Simpson, S. J., Raubenheimer, D., & Kooijman, S. A. L. M. (2013). Balancing heat, water and nutrients under environmental change: A thermodynamic niche framework (S. Chown, Ed.). *Functional Ecology*, *27*(4), 950–966. https://doi.org/10.1111/1365-2435.12020

Kooijman, S. A. L. M., Lika, K., Augustine, S., & Marn, N. (2021). Multidimensional scaling for animal traits in the context of dynamic energy budget theory. *Conservation Physiology*, *9*(1), coab086. https://doi.org/10.1093/conphys/coab086

Kooijman, S. A. L. M. (2010). *Dynamic energy budget theory for metabolic organisation* (3rd ed.). Cambridge university press.

Lika, K., Kearney, M. R., & Kooijman, S. A. (2011). The "covariation method" for estimating the parameters of the standard dynamic energy budget model II: Properties and preliminary patterns. *Journal of Sea Research*, *66*(4), 278–288. https://doi.org/10.1016/j.seares.2011.09.004

Marques, G. M., Augustine, S., Lika, K., Pecquerie, L., Domingos, T., & Kooijman, S. A. L. M. (2018). The AmP project: Comparing species on the basis of dynamic energy budget parameters (D. A. Beard, Ed.). *PLOS Computational Biology*, *14*(5), e1006100. https://doi.org/10.1371/journal.pcbi.1006100

New, M., Lister, D., Hulme, M., & Makin, I. (2002). A high-resolution data set of surface climate over global land areas. *Climate Research*, *21*(1), 1–25. https://doi.org/10.3354/cr021001

Oliveira, D. F., Sulc, A., Moan, E. L., Donati, E., Klagkou, E., Lagunes, M.-J., Dajčman, U., & Tan, T. Y. (2026). How to build Dynamic Energy Budget models: Practical guidelines for model development. *Ecological Modelling*, *521*, 111703. https://doi.org/10.1016/j.ecolmodel.2026.111703

Pecquerie, L., Petitgas, P., & Kooijman, S. A. (2009). Modeling fish growth and reproduction in the context of the dynamic energy budget theory to predict environmental impact on anchovy spawning duration. *Journal of Sea Research*, *62*(2–3), 93–105. https://doi.org/10.1016/j.seares.2009.06.002

Sousa, T., Domingos, T., & Kooijman, S. (2008). From empirical patterns to theory: A formal metabolic theory of life. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *363*(1502), 2453–2464. https://doi.org/10.1098/rstb.2007.2230

Wood, S. (2025). *Mgcv: Mixed gam computation vehicle with automatic smoothness estimation* [R package version 1.9-4]. https://doi.org/10.32614/CRAN.package.mgcv